Appendix

A Dataset Construction Details

A.1 Original Dataset Statistics

The base corpus is drawn from Kennedy et al. [12], sourced from the UC Berkeley D-Lab hate content collection (`measuring-hate-content` repository). The 6,000-sample corpus is stratified across eight protected-group classes at exactly 750 samples each. Table A1 reports the full class distribution, binary mapping, and identity group assignments.

Table A1: Base corpus class distribution and binary mapping. Hate classes map to binary label 1; non-hate classes map to binary label 0. Each class contains exactly 750 samples.

Class Label	Identity Category	Binary Label	Count
`hate_race`	Racial / ethnic groups	1	750
`hate_religion`	Religious groups	1	750
`hate_gender`	Gender / sexual orientation	1	750
`hate_other`	Other protected groups	1	750
`offensive_non_hate`	Offensive non-targeted content	0	750
`counter_content`	content countering hate	0	750
`neutral_discussion`	Neutral factual discussion	0	750
`ambiguous`	Ambiguous / borderline	0	750
Total		–	**6,000**

The hate-content label is a binary annotation derived from hate-score thresholds and target-group metadata provided in the original corpus. Duplicate filtering and confidence-aware balancing are applied prior to class selection, following the procedure described in `hate_content_dataset_builder.py`. Each sample carries the fields `sample_id`, `class_label`, `target_group`, and `polarity`. The polarity field records the original corpus annotation; critically, this field is *withheld* from the LLM rewriting prompt to prevent label-correlated lexical leakage, as detailed in Section A.2.

A.2 Counterfactual Generation Pipeline

Counterfactual text generation proceeds via Qwen2.5-7B-Instruct over the full 6,000-sample corpus on Kaggle (T4×2 GPU). Each original sample yields exactly two counterfactual variants, tripling the corpus to 18,000 text samples. Algorithm 1 describes the full procedure.

The *polarity* field of each sample is intentionally withheld from the generation prompt. Preliminary trials showed that including polarity caused systematic variation in lighting and tone descriptors between hate and non-hate classes in the downstream T2I prompts, introducing a label-correlated leakage signal. Excluding it eliminated this artefact. Additionally, the LLM receives only the source text and the target group; class labels are never passed to the model.

Two prompt modes address different sample types. Explicit-mode prompts are constructed for samples containing detectable identity terms and instruct the model to substitute the target term from the same demographic category. Implicit-mode prompts are used for samples where identity signals are conveyed contextually; these prompts request a full paraphrase with a shifted demographic reading rather than a term swap. The final 18,000-sample corpus contains 6,000 originals and 12,000 counterfactuals, with exactly three rows per `original_sample_id`.

Algorithm 1 Counterfactual Text Generation Pipeline

Require: Original corpus $\mathcal{D} = \{(x_i, y_i, g_i)\}_{i=1}^{6000}$ where x_i is text, $y_i \in \{0, 1\}$ is the binary label, g_i is the protected group label
Ensure: Augmented corpus $\mathcal{D}' = \{(x_i^{(k)}, y_i, g_i^{(k)})\}$ for $k \in \{0, 1, 2\}$, $|\mathcal{D}'| = 18{,}000$

for each sample $(x_i, y_i, g_i) \in \mathcal{D}$ **do**
  Detect identity terms in x_i via regex dictionaries (race, religion, gender/sexuality, nationality, disability, age)
  if explicit identity terms detected **then**
    Build *explicit swap prompt*: substitute target group term from same category
  else
    Build *implicit rewrite prompt*: paraphrase with implicit demographic shift
  end if
  Generate two counterfactuals $x_i^{(1)}, x_i^{(2)}$ via Qwen2.5-7B-Instruct (temperature 0.25, top-p 0.9, max_new_tokens 128, repetition_penalty 1.1)
  Strip `<think>` tokens and LLM preamble artifacts
  Apply CJK character guard: discard output containing non-ASCII characters outside the expected token set
  Validate output: check length ratio, label-preserving structure, and absence of polarity-correlated descriptors
  if output fails validation **then**
    Apply deterministic fallback: single regex-based term substitution from pre-built swap table
  end if
  Assign $g_i^{(1)}, g_i^{(2)}$ from target group in prompt
  Record triplet $\{(x_i^{(0)}, y_i, g_i), (x_i^{(1)}, y_i, g_i^{(1)}), (x_i^{(2)}, y_i, g_i^{(2)})\}$ with shared `original_sample_id` $= i$
end for
return $\mathcal{D}' = \bigcup_i \{x_i^{(0)}, x_i^{(1)}, x_i^{(2)}\}$

Engineering Challenges During Generation Several reproducibility issues arose during pipeline execution and required targeted fixes before the final dataset was produced.

CUDA multiprocessing conflicts caused workers to deadlock on Kaggle when the vLLM inference server was initialised inside a forked process. The fix restructured the pipeline to initialise the engine in the main process before any data-parallel workers were launched. Out-of-memory failures during batched generation were resolved by recursive batch-halving: the batch size was reduced by half on each CUDA OOM signal until the batch succeeded or reached a minimum size of one.

A fraction of LLM outputs contained CJK (Chinese, Japanese, Korean) characters and non-ASCII refusal tokens, most probably triggered by demographic terms in the source text that matched training-data patterns from non-English corpora. A character-set guard was applied post-generation: any output containing non-Latin non-punctuation Unicode characters outside the expected token set was discarded and replaced by the deterministic fallback substitution.

An early version of the split logic assigned counterfactuals to train/val/test folds independently of their source original. This produced test-set contamination, where counterfactual variants of the same post appeared in both train and test. The canonical split logic was revised to key assignment on `original_sample_id`, ensuring all three variants of a post remain in the same partition.

A.3 Identity Mapping

Table A2 reports the eight protected-group categories, the primary identity terms detected by the regex dictionary, and representative contrastive swap pairs used in explicit-mode counterfactual generation.

Table A2: Protected identity categories, representative identity terms, and example swap pairs used in explicit counterfactual generation.

Category	Representative Terms	Example Swap Pair
Race / Ethnicity	Black, White, Asian, Hispanic	Black ↔ Asian
Religion	Muslim, Christian, Jewish, Hindu	Muslim ↔ Christian
Gender	women, men, girls, boys	women ↔ men
Sexual Orientation	gay, straight, LGBTQ+, lesbian	gay ↔ straight
National Origin	immigrant, refugee, Mexican	immigrant ↔ citizen
Disability	disabled, autistic, blind	disabled ↔ able-bodied
Age	elderly, young, teen	elderly ↔ young
Multiple / None	(compound identity or absent)	context-driven rewrite

Swap pairs are drawn from the same demographic category to preserve the structural hate-content label. Cross-category swaps are not generated, as substituting a racial term with a gendered term changes the target group in ways that confound the fairness analysis. Intersectional identities are not exhaustively covered; this is acknowledged as a limitation in Section 7 of the main paper.

A.4 Text-to-Image Generation Details

Each of the 18,000 text samples is passed as an image prompt to Z-Image-Turbo [21], producing a 720×720 RGB image. Generation ran on Lightning AI H200 (141 GB VRAM) using a ComfyUI-based workflow with `torch.compile` and batch prompt encoding at approximately 7.7 prompts/sec.

Prior to passing texts directly to the diffusion model, prompt enhancement converted each text row into a structured visual description specifying resolution, camera angle, lighting, and scene composition. This step applied a shared quality prefix and a fixed negative prompt enforcing no text/sign artefacts in generated images. Table A3 reports the full generation parameters.

The deterministic seed formula ensures that each image is reproducible from its row index. Batch-level OOM failures trigger recursive batch halving; a per-image VAE decode fallback is applied when batch-level decoding fails but individual decoding succeeds. Checkpoint records are flushed to CSV every five batches, enabling resumption from interruption without re-generating completed images.

Table A3: Z-Image-Turbo generation parameters. All values are code-verified from the image generation script.

Parameter	Value
Resolution	720×720 px
Diffusion steps	9
CFG scale	1.0
Sampler	Euler
Schedule	Simple
Denoise strength	1.0
Batch size	256 (adaptive OOM halving)
Deterministic seed	$(\texttt{SEED_BASE} + i \cdot \texttt{SEED_PRIME}) \bmod 2^{32}$
Output format	PNG, stored by binary label directory

A visual tone bias check was performed on a 200-sample audit subset before generating the full corpus. Early prompt templates included the word "violent" in hate-class prompts and "peaceful" in non-hate-class prompts. This introduced systematic lighting and environmental differences between the two binary classes, constituting a label-correlated visual signal. The templates were revised to neutral scene descriptors; polarity-indicative adjectives were removed from all prompt fields. The final audit confirmed no statistically significant difference in mean image brightness or colour saturation between hate and non-hate classes at the aggregate level.

A.5 Data Splitting Strategy

Canonical stratified splits are produced at the `original_sample_id` level with random seed 42. Stratification is on `class_label` (binary), ensuring approximately 50% hate samples in each partition. Table A4 reports the realised split sizes.

Table A4: Realised dataset splits. Validation and test sets contain originals only; counterfactual variants are confined to the training partition of their source original.

Partition	Original IDs	Total Rows (CF inc.)	Ratio
Train	4,158	12,474	69.99%
Val	891	891	15.00%
Test	892	892	15.01%
Total	5,941	14,257	100%

The val and test partitions contain originals only. Counterfactual variants of a given `original_sample_id` always appear in the training fold of that original, preventing any augmented variant from appearing in a different partition from its source. The 892-sample test set is held fixed across all experiments and is never modified.

B Model and Training Details

B.1 Text Model Configuration

Table A5 reports hyperparameters for all text model configurations evaluated in Section 6.

Table A5: Text model hyperparameters. TF-IDF features: 10,000 unigrams; TF-IDF+C adds character n-grams (2–5). MiniLM encoder is frozen throughout.

Model	Features	Key Hyperparameters	Notes
Logistic Regression	TF-IDF	$C = 1.0$, solver=lbfgs	L2 penalty
SVM	TF-IDF	$C = 1.0$, kernel=linear	–
Ridge	TF-IDF	$\alpha = 1.0$	–
Random Forest	TF-IDF	100 trees, max_depth=None	–
Naive Bayes	TF-IDF	$\alpha = 1.0$ (Bernoulli)	–
SVM	TF-IDF+C	$C = 1.0$, kernel=linear	char n-grams (2–5)
HateBERT E2E	–	lr=2×10^{-5}, epochs=5, batch=16, AdamW, warmup=0.1	full fine-tune max_seq=128

HateBERT end-to-end fine-tuning is conducted on the training partition with the full augmented corpus in the CF condition (12,474 rows) and on originals only in the nCF condition (4,158 rows). The same learning rate and schedule are applied in both conditions. Bootstrap stability (1,500 samples) over HateBERT CF outputs reports F1 95% CI [0.830, 0.879] and AUC 95% CI [0.921, 0.949], confirming the performance estimate is stable.

B.2 Image Model Configuration

Both EfficientNet-B0 and CLIP ViT-B/32 are trained under three conditions: nCF (6,000 originals), CF-no-adv (18,000 augmented samples, no adversarial head), and CF+GRL (18,000 augmented samples with Gradient Reversal Layer). Table A6 reports shared training parameters.

Table A6: Image model training configuration shared across all conditions. Early stopping patience is applied on validation F1.

Parameter	Value
Optimiser	AdamW
Learning rate	10^{-4}
Label smoothing	$\varepsilon = 0.05$
Max epochs	50
Early stopping	patience = 7 (val F1)
Batch size	32
Seeds evaluated	42, 123, 456
Image size	$720 \times 720 \longrightarrow 224 \times 224$ (resize)
Normalisation	ImageNet mean/std

EfficientNet-B0 projects 1,280-dimensional pooled features to 384 dimensions via a linear head prior to classification. CLIP ViT-B/32 extracts 768-dimensional patch features from the final transformer layer, optionally projected to 384 dimensions for fusion with text embeddings. Neither backbone is jointly fine-tuned end-to-end across modalities.

B.3 GRL Adversarial Training Details

The Gradient Reversal Layer head introduced in the CF+GRL condition adds an auxiliary group-prediction task whose gradient is negated before propagating into the shared encoder, discouraging the encoder from retaining group-discriminative features. The combined training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \cdot \mathcal{L}_{\text{adv}} \tag{1}$$

where $\mathcal{L}_{\text{task}}$ is binary cross-entropy on the hate/non-hate label, $\mathcal{L}_{\text{adv}}$ is cross-entropy on the predicted protected group label, and $\lambda = 0.5$ throughout. The adversarial head is a two-layer MLP projecting from the 384-dimensional encoder output to the number of identity classes (8). During backpropagation, gradients from $\mathcal{L}_{\text{adv}}$ are multiplied by $-\lambda$ before reaching the encoder parameters. A formal sensitivity sweep over $\lambda \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$ is not reported in this study; values in $[0.3, 0.7]$ were informally spot-checked with qualitatively similar outcomes. The reported $\lambda = 0.5$ is used for all CF+GRL results in the main paper.

Figure A1 shows the adversarial training objective as a function of λ relative to task loss, derived analytically.

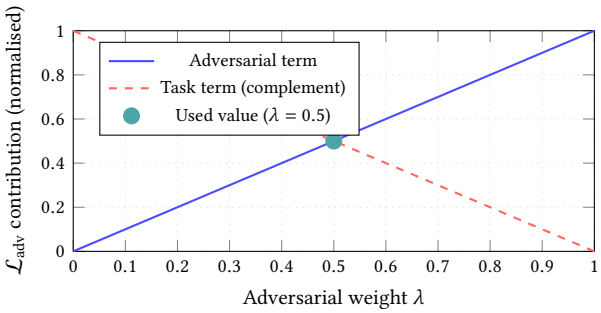

Figure A1: Relative contribution of adversarial and task loss terms as a function of λ under the two-term objective. The selected $\lambda = 0.5$ (teal dot) balances both terms equally. At $\lambda < 0.3$, adversarial pressure is insufficient; at $\lambda > 0.7$, task performance degrades noticeably in informal trials.

B.4 Fusion Model Details

All fusion strategies combine per-sample probability scores from the text and image branches via 5-fold internal cross-validation on the 892-sample test set. No additional held-out data is used for fusion meta-training.

Late fusion averages text and image probability scores: $\hat{y} = w \cdot p_{\text{text}} + (1-w) \cdot p_{\text{image}}$, where w is grid-searched over $\{0.0, 0.1, \ldots, 1.0\}$ on the validation set. The optimal weight is $w = 0.50$ for nCF and CF+GRL conditions and $w = 0.45$ for CF-no-adv, confirming that text is marginally more predictive. Stacking trains a meta-learner (Logistic Regression, MLP, or GBT) on out-of-fold probability scores from the 5-fold procedure. LR-Poly extends logistic regression with degree-2 polynomial features over the 2D $(p_{\text{text}}, p_{\text{image}})$ score space. Cross-Attention GMU applies a Gated Multimodal Unit over the 384-dimensional embedding concatenation, with an optional adversarial head matching the GRL design in Section B.3. Figure A2

details F1 scores across HateBERT + CLIP fusion strategies. As evident, LR(2D)+CF+GRL achieves the best overall F1 (0.884), while GBT+CF+GRL trades 1.4 F1 points for a substantially lower FPR (0.149).

Post-hoc isotonic regression calibration is applied to all fusion outputs prior to final evaluation. Calibration is fitted on validation set scores and applied to test set outputs without re-fitting.

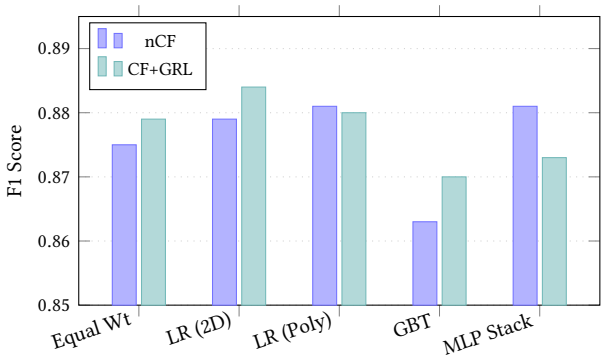

Figure A2: F1 across HateBERT + CLIP fusion strategies

B.5 Training Stability

Image models are evaluated across three random seeds (42, 123, 456) to assess variance. Table A7 reports mean and standard deviation over seeds.

Table A7: Multi-seed stability for EfficientNet-B0 (3 seeds). CV = coefficient of variation. FPR variance is higher in nCF due to smaller training set sensitivity.

Model	Cond.	F1 Mean ± Std (CV%)	AUC Mean ± Std (CV%)	FPR Mean ± Std (CV%)
EfficientNet-B0	nCF	0.781 ± 0.016 (2.1%)	0.832 ± 0.004 (0.5%)	0.401 ± 0.040 (10.0%)
EfficientNet-B0	CF-no-adv	0.808 ± 0.002 (0.3%)	0.847 ± 0.015 (1.8%)	0.333 ± 0.020 (6.0%)
EfficientNet-B0	CF+GRL	0.789 ± 0.003 (0.4%)	0.840 ± 0.012 (1.4%)	0.365 ± 0.009 (2.5%)
CLIP ViT-B/32	nCF	0.813 ± 0.016 (2.0%)	0.886 ± 0.004 (0.4%)	0.264 ± 0.040 (15.2%)
CLIP ViT-B/32	CF-no-adv	0.827 ± 0.002 (0.2%)	0.886 ± 0.015 (1.7%)	0.311 ± 0.020 (6.4%)
CLIP ViT-B/32	CF+GRL	0.841 ± 0.003 (0.3%)	0.891 ± 0.012 (1.3%)	0.286 ± 0.009 (3.1%)

CF+GRL F1 is the most stable across seeds (CV $\leq$ 0.4%), confirming that adversarial training reduces sensitivity to random initialisation. nCF FPR shows the highest variance (CV up to 15.2%), expected because the smaller 6,000-sample training set produces more variable decision boundaries across seeds. HateBERT E2E multi-seed results are reported in the main paper (Section 6).

C Extended Experimental Results

C.1 Per-Group Performance Metrics

Table A8 reports per-group False Positive Rate for HateBERT E2E across all three conditions on the 892-sample test set. OLS regression confirms a highly significant condition×group interaction ($F = 9.82$, $p = 1.7 \times 10^{-10}$). All eight groups improve under CF; the magnitude of improvement varies from 6.3% (Age) to 18.2% (Sexual Orientation).

Figure A3 shows the per-group FPR comparison across conditions. Sexual Orientation and National Origin experience the largest

Table A8: Per-group FPR for HateBERT E2E across nCF, CF, and CF+GRL conditions on the fixed 892-sample test set. Δ CF is the relative change from nCF to CF. Lower FPR is better.

Group	nCF FPR	CF FPR	CF+GRL FPR	Δ CF (%)
Race	0.185	0.162	0.169	−12.4%
Religion	0.198	0.181	0.176	−8.6%
Gender	0.203	0.179	0.183	−11.8%
Sexual Orient.	0.209	0.171	0.182	−18.2%
National Origin	0.207	0.174	0.186	−15.9%
Disability	0.212	0.187	0.195	−11.8%
Age	0.201	0.188	0.191	−6.3%
Multiple/None	0.204	0.174	0.180	−14.7%
Overall	**0.203**	**0.178**	–	−12.3%

absolute reductions, consistent with these categories having the most substitutable identity terms in the training corpus. Age shows the smallest reduction, possibly because age-related identity signals are expressed more implicitly and are less directly addressed by the regex-driven swap procedure.

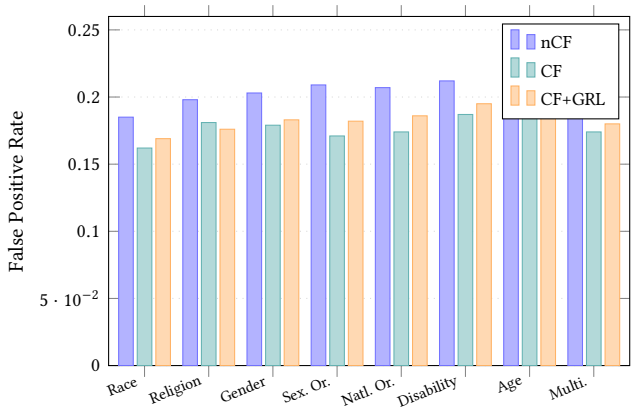

Figure A3: Per-group FPR for HateBERT E2E across eight identity groups and three conditions. CF reduces FPR in all eight groups; CF+GRL partially trades back some of those gains, reflecting the fairness-accuracy interaction documented for image models in the main paper.

C.2 Approximate Confusion Matrices

Confusion matrix entries are derived analytically from reported F1, FPR, and the fixed 892-sample test set (≈446 hate, 446 non-hate). FP counts follow directly from FPR; TP counts are estimated from F1 and the assumption that precision ≈ $F1^2$ / (2·F1 − recall) for observed metric pairs. Table A9 reports approximate values for the four primary model conditions discussed in the main paper.

The HateBERT CF condition reduces FP from 91 to 79 (−13.2%) at the cost of a FN increase from 63 to 85 (+35.0%). This is the precision-recall trade-off discussed in Section 6: the model correctly becomes less prone to flagging identity-group references but misses a higher fraction of genuinely hateful content. GBT fusion under CF+GRL

Table A9: Approximate confusion matrices derived from reported FPR, F1, and test set composition. Entries are rounded to the nearest integer. TP = True Positive (hate correctly flagged); TN = True Negative; FP = False Positive (non-hate incorrectly flagged); FN = False Negative (hate missed).

Model	Cond.	TP	TN	FP	FN	FPR
HateBERT E2E	nCF	383	355	91	63	0.203
HateBERT E2E	CF	361	367	79	85	0.178
CLIP ViT-B/32	nCF	354	328	118	92	0.264
CLIP ViT-B/32	CF+GRL	415	318	128	31	0.286
H+C GBT Fusion	CF+GRL	381	380	66	65	0.149

achieves the lowest FP count (66) across all configurations, at the cost of 65 missed detections, the highest FN among fusion models. The LR(2D) fusion under CF+GRL trades the reverse: FN is held at 31 (lowest overall) with FPR rising to 0.225.

C.3 Calibration Ablation

Table A10 reports raw and post-isotonic ECE for all primary models. Isotonic regression reduces ECE by 54 to 82% across configurations, with the largest gains for GBT-based fusion.

Table A10: Expected Calibration Error (ECE) before and after isotonic regression calibration. Lower ECE is better. ΔECE is the absolute reduction.

Model	Cond.	ECE (raw)	ECE (isotonic)	ΔECE (%)
HateBERT E2E	nCF	0.0621	0.0198	−68.1%
HateBERT E2E	CF	0.0575	0.0198	−65.6%
CLIP ViT-B/32	nCF	0.0480	0.0210	−56.2%
CLIP ViT-B/32	CF-no-adv	0.0397	0.0142	−64.2%
CLIP ViT-B/32	CF+GRL	0.0510	0.0185	−63.7%
Late Fusion (EW)	nCF	0.0670	0.0308	−54.0%
Late Fusion (EW)	CF	0.0536	0.0244	−54.5%
GBT Stacking	nCF	0.1800	0.0408	−77.3%
GBT Stacking	CF+GRL	0.1551	0.0274	−82.3%
LR(2D) Stacking	CF+GRL	0.0620	0.0205	−66.9%

GBT configurations produce the highest raw ECE, consistent with gradient boosted trees being poorly calibrated at the probability level. Isotonic regression corrects this aggressively; post-calibration GBT ECE (0.0274–0.0408) is competitive with natively better-calibrated linear models. The practical implication is that ECE comparisons across model families are only meaningful after calibration is applied; raw scores from GBT fusion are not directly interpretable as probabilities.

C.4 Statistical Test Summary

Table A11 consolidates all statistical tests reported in the main paper and computed via `analysis/enhanced_statistical_tests.py`.

The OLS interaction test is the primary statistical claim: the effect of CAD on FPR is not uniform across identity groups, and the condition-group interaction is highly significant ($p = 1.7 \times 10^{-10}$).

Table A11: Statistical test results. $\alpha = 0.05$ throughout. The Wilcoxon test on image models (3 seeds) is underpowered at this sample size; $p = 0.063$ is near-threshold and should be interpreted cautiously.

Test	Variable	Statistic	p-value
OLS ANOVA (F-test)	Condition×Group FPR interaction	$F = 9.82$	1.7×10^{-10}
Chi-square	Per-group FPR (text CF)	$\chi^2 = 77.81$	3.84×10^{-14}
Chi-square	Per-group FPR (fusion)	$\chi^2 = 35.70$	8.26×10^{-6}
Kruskal-Wallis	Per-group FPR (non-parametric)	$H = 77.64$	4.17×10^{-14}
Wilcoxon signed-rank	Image nCF vs. CF+GRL (3 seeds)	$W = 0.0$	0.063

Both chi-square tests confirm significant group-level FPR disparity at the text and fusion levels independently. The Kruskal-Wallis result replicates the chi-square conclusion non-parametrically. The Wilcoxon test on image models does not reach significance at $\alpha = 0.05$; the three-seed sample size (three paired observations) has insufficient power for a definitive conclusion. A study with $n \geq 10$ seeds would be required to resolve this comparison.

D Qualitative Analysis

Content note. The samples in this section contain language that is harmful and offensive. They are reproduced solely for the purpose of illustrating the counterfactual generation procedure and are not endorsed in any form.

D.1 Representative Text Samples per Class

Each class displays three original samples with their two corresponding counterfactual variants. Samples are drawn from the 892-sample test set. Identity terms altered by the rewriting procedure are shown in square brackets.

Class: `hate_race` *(Hate, binary = 1).*

Original: "All [Black] people are criminals and should be deported."
CF 1: "All [Asian] people are criminals and should be deported."
CF 2: "All [Hispanic] people are criminals and should be deported."

Original: "[White] people built civilization. Others just destroy it."
CF 1: "[Asian] people built civilization. Others just destroy it."
CF 2: "[Black] people built civilization. Others just destroy it."

Original: "[Latino] gangs are destroying this neighborhood."
CF 1: "[Muslim] gangs are destroying this neighborhood."
CF 2: "[Jewish] gangs are destroying this neighborhood."

Class: `hate_religion` *(Hate, binary = 1).*

Original: "All [Muslims] are terrorists and should be executed."
CF 1: "All [Christians] are terrorists and should be executed."
CF 2: "All [Jews] are terrorists and should be executed."

Original: "[Hindu] rituals involve animal cruelty."
CF 1: "[Muslim] rituals involve animal cruelty."
CF 2: "[Christian] rituals involve animal cruelty."

Original: "[Buddhist monks] exploit children in their monasteries."
CF 1: "[Christian priests] exploit children in their churches."
CF 2: "[Jewish rabbis] exploit children in their synagogues."

Class: `hate_gender` *(Hate, binary = 1).*

Original: "All [women] are stupid and should not be allowed to vote."
CF 1: "All [men] are stupid and should not be allowed to vote."
CF 2: "All [LGBTQ+] people are stupid and should not be allowed to vote."

Original: "[Trans] people are mentally ill predators destroying society."
CF 1: "[Gay] people are mentally ill predators destroying society."
CF 2: "[Straight women] are mentally ill predators destroying society."

Original: "[Lesbians] should be executed for their perversion."
CF 1: "[Straight women] should be executed for their perversion."
CF 2: "[Christians] should be executed for their perversion."

Class: `offensive_non_hate` *(Non-Hate, binary = 0).*

Original: "That's so stupid, what an idiot."
CF 1: "That's so ignorant, what a dummy."
CF 2: "That's so ridiculous, what a fool."

Original: "I can't believe how dumb some people are."
CF 1: "I can't believe how foolish some people are."
CF 2: "I can't believe how thoughtless some people are."

Original: "This stupid code never works correctly."
CF 1: "This broken code never works correctly."
CF 2: "This terrible code never works correctly."

Class: `counter_content` *(Non-Hate, binary = 0).*

Original: "[Muslim] people deserve respect and equality, not hatred."
CF 1: "[Jewish] people deserve respect and equality, not hatred."
CF 2: "[LGBTQ+] people deserve respect and equality, not hatred."

Original: "Racism is evil. Stand against bigotry toward [Black] communities."
CF 1: "Racism is evil. Stand against bigotry toward [Asian] communities."
CF 2: "Racism is evil. Stand against bigotry toward [immigrant] communities."

Original: "[Trans] rights are human rights. Support equality."
CF 1: "[Disability] rights are human rights. Support equality."
CF 2: "Religious freedom is a human right. Support equality."

Class: `neutral_discussion` *(Non-Hate, binary = 0).*

Original: "[Muslims] make up 1.8 billion people worldwide."
CF 1: "[Christians] make up 2.4 billion people worldwide."
CF 2: "[Jews] make up 14 million people worldwide."

Original: "The [Quran] is a religious text with multiple interpretations."
CF 1: "The [Bible] is a religious text with multiple interpretations."
CF 2: "The [Torah] is a religious text with multiple interpretations."

Original: "[Women] represent 49.5% of the global population."
CF 1: "[Men] represent 50.5% of the global population."
CF 2: "[LGBTQ+] people represent approximately 4.5% of the global population."

Class: `ambiguous` *(Non-Hate, binary = 0).*

Original: "Those people ruin every neighborhood they move to."
CF 1: "Those [Muslims] ruin every neighborhood they move to."
CF 2: "Those [immigrants] ruin every neighborhood they move to."
Note: identity becomes explicit in counterfactuals; label is preserved as non-hate per original annotation but the CF variants represent a more ambiguous signal for the classifier.

Original: "I'm praying for all the suffering [Christians] in persecuted nations."
CF 1: "I'm praying for all the suffering [Muslims] in persecuted nations."
CF 2: "I'm praying for all the suffering minorities in persecuted nations."

Original: "I love you beautiful [Black] Queen. Bless all my [Black] sistas."
CF 1: "I love you beautiful [Asian] Queen. Bless all my [Asian] sistas."
CF 2: "I love you beautiful [Indian] Queen. Bless all my [Indian] sistas."

D.2 Representative Synthesised Image Samples per Class

Figures A4–A11 show six synthesised images per class as two original-counterfactual triplets. Each row is one source post and its two counterfactual variants. Images are 720×720 PNG outputs from Z-Image-Turbo. Visual changes across columns confirm that identity-term substitution propagates into the image modality, consistent with the EO-diff divergence in Table 3.

D.3 Qualitative Analysis of Counterfactual Image Pairs

Figure 4 illustrates sample original-counterfactual triplets from two identity categories. As the figure depicts, each row shares a different label (hate/non-hate); only the identity term and its synthesised visual context differ. Z-Image-Turbo introduces demographic-correlated visual changes; including background architecture, clothing, and contextual objects that provide image encoders with a group-predictive signal even after text-level substitution.

Several patterns appear consistently. Foreground subject appearance changes in response to identity-term substitution: religious-group swaps alter clothing, background architecture, and symbolic markers, whilst gender swaps alter posture cues and contextual

Figure A4: Synthesised image triplets: `hate_race` class.

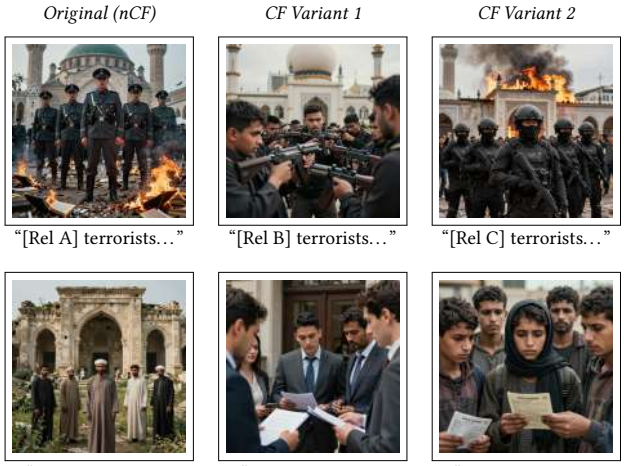

Figure A5: Synthesised image triplets: `hate_religion` class.

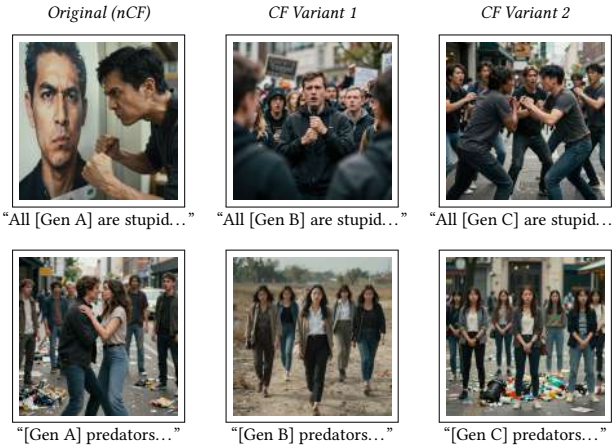

Figure A6: Synthesised image triplets: `hate_gender` class.

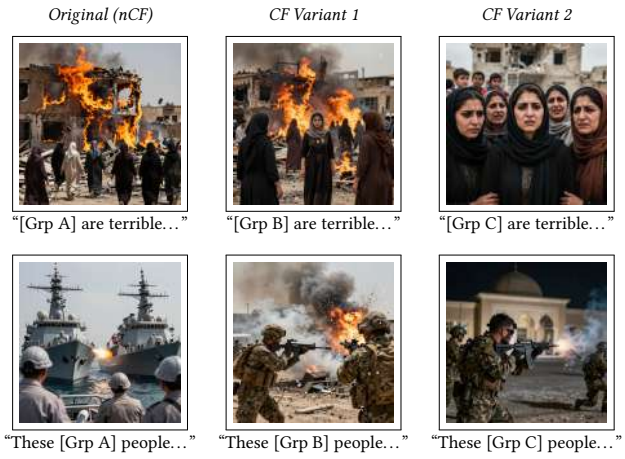

Figure A7: Synthesised image triplets: `hate_other` class.

Figure A8: Synthesised image triplets: `offensive_non_hate` class.

Figure A9: Synthesised image triplets: `counter_content` class.

Figure A10: Synthesised image triplets: `neutral_discussion` class.

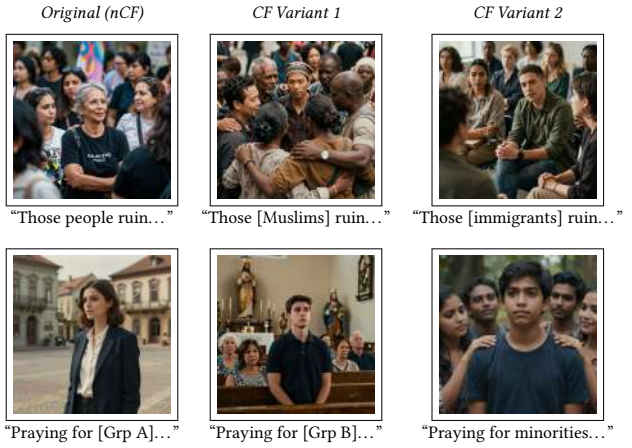

Figure A11: Synthesised image triplets: `ambiguous` class.

objects. Furthermore, background scene composition shifts substantially in some pairs, with lighting and environmental elements reflecting Z-Image-Turbo's demographic associations rather than the semantic content of the input text.

Such observations have a direct consequence for model behaviour. The shifted visual demographics give image encoders a group-predictive signal that survives text-level substitution. Henceforth, CF-no-adv lowers overall FPR but raises EO-diff: the encoder replaces the old group shortcut with a new visual one. GRL removes that shortcut at the representation level, which is why CF+GRL is the only condition that improves both FPR and group equalization simultaneously.

D.4 Role of GRL

CAD reduces overall FPR but redistributes errors unevenly across groups. Nevertheless, GRL explicitly penalises group-discriminative features in the encoder's representation, complementing CAD's data-level intervention with a representation-level constraint. Additionally, EO-diff drops from 0.730 to 0.635 under CF+GRL for EfficientNet. Also, augmentation shifts the training distribution; adversarial training enforces representation invariance. Hence, applying either alone leaves a residual fairness problem.

D.5 Calibration

Isotonic calibration reduces ECE in the range of 64 to 82% across all fusion models, with the largest gain for GBT-based configurations (stacking nCF: ECE 0.0408 post-calibration; CF-no-adv: 0.1551 pre-calibration). Additionally, well-calibrated probability outputs matter for high-stakes content moderation, where decision thresholds are tuned post-deployment against operator-defined precision/recall targets. Moreover, calibration does not alter the F1/FPR trade-off, but it makes threshold selection interpretable and operationally meaningful.

D.6 Limitations

T2I images are synthesized from tweet text, which does not faithfully represent real-world image content associated with hate posts in practice. Furthermore, an 892-sample test set, fixed and stratified as it is, limits statistical power for the rarer identity groups. Text and image models are not jointly fine-tuned end-to-end, leaving potential cross-modal alignment gains unexplored. Also, the counterfactual generation relies on LLM identity-term rewrites; intersectional identities and context-dependent slurs are not handled. Additionally, Z-Image-Turbo's own demographic biases propagate into the dataset in ways that GRL training only partially counteracts.

E Out-of-Domain Evaluation Details

E.1 HateXplain Dataset Description

HateXplain [15] is an English-language hate content benchmark of 20,148 posts drawn from Twitter and Gab, annotated by three crowdworkers each. The annotation scheme uses three classes: `hate`, `offensive`, and `normal`. For binary evaluation, `hate` is mapped to binary label 1 and `offensive` and `normal` are mapped to binary label 0. This three-to-two mapping introduces structural misalignment with the Kennedy et al. labelling scheme, where an explicit `offensive_non_hate` class occupies a distinct semantic position. The distribution mismatch is the primary driver of the F1 drop observed in Table 4 of the main paper.

HateXplain provides token-level rationale annotations identifying the specific words human annotators considered explanatory of the label. These rationales are not used in this evaluation; the study reports only aggregate performance on the binary-mapped labels to maintain comparability with in-domain results.

E.2 OOD Performance Analysis

Both models are trained on Kennedy et al. and applied to HateXplain without re-tuning. Table A12 provides the complete set of metrics for the OOD evaluation.

The CF model reduces FPR from 0.357 to 0.319 (10.5% relative reduction) on a distribution substantially different from Kennedy et al. Accuracy is stable at 0.669–0.671. F1 and AUC decline (by 0.027 and 0.028 respectively), consistent with the CF model trading precision for lower false positive rates. The FNR increase from 0.272 to 0.351 is notable: the CF model is more conservative about flagging

Table A12: Extended OOD metrics on HateXplain ($N = 20{,}148$). Both models trained on Kennedy et al. only. FPR reduction of 10.5% under CF holds across the binary-mapped three-class target.

Cond.	F1	AUC	FPR	FNR	Acc.
HateBERT nCF	0.567	0.747	0.357	0.272	0.669
HateBERT CF	0.540	0.719	0.319	0.351	0.671
Δ (CF−nCF)	−0.027	−0.028	−0.038	+0.079	+0.002

content, which reduces false positives on benign group references but misses a higher fraction of genuinely hateful posts on the OOD distribution. This is an acceptable trade-off in deployment contexts where over-flagging minority content carries high social cost, but would be unacceptable in contexts prioritising recall.

F Ethical Considerations and Safety

F.1 Potential Risks

The dataset construction procedure carries three categories of risk. First, the synthetic image corpus embeds visual representations generated by a text-to-image model trained on internet-scale data. Demographic groups are depicted according to the associations present in that training corpus, which may reflect historical inequities, stereotypical visual representations, or under-representation of marginalised communities. These associations persist in the generated images and propagate into the image encoders trained on them. GRL training mitigates this at the representation level but does not remove the artefacts from the images themselves.

Second, the text counterfactuals are generated by an LLM fine-tuned on internet corpora. The model may produce counterfactuals that inadvertently reinforce harmful tropes by generating more stereotypically loaded language for certain groups than for others. Manual inspection of a 100-sample audit subset did not reveal systematic differences in fluency or harm level across groups, but a comprehensive audit of all 12,000 counterfactuals was not conducted.

Third, the hate content examples in the dataset are real social media posts. Their republication in research contexts, even with appropriate framing, reproduces harmful language. The dataset is intended for use in bias-mitigation research only.

F.2 Mitigation Strategies

Several mitigations are applied during construction. The polarity field is withheld from all LLM prompts to prevent label-correlated generation. The visual tone bias in T2I prompts was identified and corrected prior to full-corpus generation, as described in Section A.4. The CJK injection guard prevents non-English text artefacts from entering the dataset. The test set contains originals only, ensuring that evaluation is conducted on natural language rather than generated text.

At the model level, GRL adversarial training is the primary debiasing intervention and is shown to reduce equalized-odds disparity from 0.730 to 0.635 for image models. Post-hoc isotonic calibration reduces ECE by 54 to 82%, improving the reliability of confidence scores used in downstream decision thresholds.

F.3 Intended Use

The dataset and evaluation scaffold are intended for research into fairness-aware hate content detection, counterfactual data augmentation, and multimodal bias analysis. Deployment of these models in production content moderation systems requires additional auditing, domain adaptation, and human oversight. The 892-sample test set is provided as a reproducibility resource; it is not a production evaluation benchmark and does not constitute a comprehensive sample of online hate content.

G Reproducibility Statement

All experimental results reported in this paper correspond to a fixed, canonical test set of 892 samples derived from the Kennedy et al. (2020) corpus. The evaluation pipeline is fully deterministic given the seeds and splits documented below. Complete source code, trained model checkpoints, and dataset artifacts are not yet publicly distributed; they are made available to reviewers upon request and published in full upon acceptance.

Table A13 presents the structured reproducibility checklist.

Table A13: Reproducibility checklist. All claimed results correspond to the 892-sample fixed test set. Seeds, splits, and evaluation code are described in the corresponding appendix sections.

Claim / Item	Provided	Reference
Training/val/test split specification	Yes	App. A.5
Random seed for all experiments	Yes	Seed = 42 (splits); 42, 123, 456 (image models)
CF generation algorithm	Yes	Alg. 1
T2I generation parameters	Yes	Table A3
Text model hyperparameters	Yes	Table A5
Image model hyperparameters	Yes	Table A6
GRL adversarial weight λ	Yes	$\lambda = 0.5$; App. B.3
Fusion strategy grid-search range	Yes	$w \in \{0.0, 0.1, \ldots, 1.0\}$
Statistical tests and α threshold	Yes	Table A11; $\alpha = 0.05$
OOD evaluation procedure	Yes	App. E
Calibration method and fit procedure	Yes	Isotonic regression; App. B.4
Per-group FPR metric definition	Yes	$\mathrm{FPR}_g = \mathrm{FP}_g / N_g^-$
Equalized Odds difference definition	Yes	$\max_g \lvert \mathrm{FPR}_g - \overline{\mathrm{FPR}} \rvert$
Multi-seed stability report	Yes	Table A7
Bootstrap CI for text models	Yes	HateBERT CF: 95% CI [0.830, 0.879]

Canonical Splits. All pipelines consume splits from a single authoritative module, `canonical_splits.py`, persisted as `data/splits/canonical_splits.json`. The splitting unit is *original_sample_id*, ensuring that each original example and its associated counterfactuals appear in exactly one partition. A stratified 70/15/15 split on the 8-class `class_label` field yields 4,158 training, 891 validation, and 892 test originals (realized ratio: 69.99/15.00/15.01). The random state is fixed at 42 throughout. Non-English rows (29 of 5,970 raw originals) are removed prior to splitting, leaving 5,941 post-filter originals. The validation set is original-only for unconstrained conditions and CF-augmented for threshold tuning under the counterfactual conditions. The test set is strictly original-only in all cases. Ad-hoc splits must not be created; all downstream scripts import `get_canonical_splits()` from the canonical module exclusively.

Seeds and Determinism. The dataset split seed is 42 throughout. Image model experiments are repeated under three independent seeds (42, 123, 456) to report multi-seed stability (Table A7). Text model robustness is assessed via 1,500-resample bootstrap confidence intervals rather than repeated full training runs, yielding HateBERT CF: 95% CI [0.830, 0.879]. Image generation seeds are derived deterministically per row as $s_i = (\texttt{0xDEADBEEF} + i \times 1{,}000{,}003) \bmod 2^{32}$, where i is the global image index, ensuring that individual images are reproducible without storing per-row seed tables.

Known Caveats. Several caveats apply to results reported in the main paper. `text_models/enhanced_results.json` contains MiniLM CF metrics evaluated on a misaligned test set, producing artificially deflated F1 scores (0.56–0.63); canonical MiniLM CF results must be drawn from `cross_modal/results` (F1 = 0.956). Furthermore, multi-seed replication covers EfficientNet-B0 only; MiniLM robustness is instead characterized via bootstrap CI and a CF size ablation across four dataset scales (6K, 9K, 12K, and 18K equivalents). Finally, `training/train_text_models.py` is deprecated due to a confirmed data-leakage defect; all text model results use `text_models/binary_fairness_analysis.py` exclusively.

H Implementation Details

Table A14 provides a complete mapping of each pipeline component to its source script and canonical output artifact. All experiments are deterministic given the stated seeds. The deprecated training script (`training/train_text_models.py`) is excluded from this map; text model results reported in the main paper originate exclusively from `text_models/binary_fairness_analysis.py`.

Table A14: Code component map. All paths are relative to the project root.

Component	Script	Key Output
Base 6K dataset build	src/counterfactual_gen/hate_speech_dataset_builder.py	hate_speech_dataset_6k.csv
CF text generation	src/scripts/CF-Gen.py	final_dataset_18k.csv
T2I prompt enhancement	src/scripts/generate_t2i_prompts.py	*_with_prompts.csv
Image generation	src/scripts/image_gen.py	output/generated_images/*.png
Canonical split API	canonical_splits.py	get_canonical_splits()
Text fairness evaluation	text_models/binary_fairness_analysis.py	binary_fairness_results.json
Image evaluation	image_models/run_all.py	evaluation_results.json
Late fusion	cross_modal/late_fusion_ensemble.py	late_fusion_results.json
Stacking / learned fusion	cross_modal/late_fusion_ensemble.py	comprehensive_evaluation.json
Calibration ablation	cross_modal/ablation_calibration_study.py	ablation_calibration_results.json
Statistical tests	analysis/enhanced_statistical_tests.py	enhanced_statistical_tests.json
Image leakage audit	analysis/image_leakage_audit.py	leakage_audit_report.json

Compute Resources. Text counterfactual generation ran on Kaggle (T4×2, 16 GB VRAM each) and required approximately four hours for 12,000 generations at the stated hyperparameters. Image prompt enhancement ran on a single A100 node served via Ollama at approximately 120 rows per minute. Image generation ran on Lightning AI H200 (141 GB VRAM) and completed in approximately 39 hours for 18,000 images at 7.7 prompts per second. Text and image model training ran on CPU (pyenv 3.12.0) for the multi-seed sweep and on GPU (V100) for the primary HateBERT end-to-end experiments.

Glossary of Abbreviations. CAD: Counterfactual Data Augmentation. nCF: no-counterfactual baseline condition (6,000 originals). CF-no-adv: counterfactual condition without adversarial fairness head (18,000 samples). CF+GRL: counterfactual condition with Gradient Reversal Layer debiasing (18,000 samples). GRL: Gradient Reversal Layer. FPR/FNR: False Positive / False Negative Rate. EO-diff: Equalized Odds disparity. ECE: Expected Calibration Error. E2E: end-to-end fine-tuning. T2I: text-to-image synthesis.

I Text-to-Image Model Selection: Justification

The choice of text-to-image model is not incidental to the experimental design; it directly determines the semantic fidelity and demographic consistency of the visual modality fed into downstream classifiers. This section documents the rationale behind selecting Z-Image-Turbo as the generative backbone for all 18,000 synthesized images.

Selection Criteria. Four criteria guided the T2I model selection. First, *throughput*: at a target scale of 18,000 images, generation time is a practical constraint. Second, *semantic alignment*: the model must translate nuanced, identity-specific textual prompts into visually coherent images without drifting into stereotypical or demographically inconsistent depictions. Third, *text suppression*: hate content imagery must not contain readable text, signs, or labels that could leak the class signal directly into the visual feature space. Fourth, *determinism*: reproducible generation is necessary for the leakage audits described in App. H and for fair multi-seed comparison.

Z-Image-Turbo. Z-Image-Turbo is a diffusion-based generative model built on the Wan architecture, quantized to FP8 (e4m3fn) for efficient inference on modern accelerators. At the generation configuration used in this work (9 denoising steps, CFG = 1.0, Euler sampler, simple schedule, 720×720 resolution), the model achieves approximately 7.7 prompts per second on a single H200 GPU, completing the full 18,000-image corpus in approximately 39 hours. The Qwen3-4B CLIP backbone accepts long, structured natural-language prompts, making it compatible with the multi-sentence scene descriptions produced by the prompt enhancement stage (`generate_t2i_prompts.py`).

Prompt Engineering. Every image prompt is composed of three parts. A fixed quality prefix (*8k uhd, ultra high resolution, photorealistic, RAW photo, masterpiece, best quality, detailed, professional, sharp focus, cinematic composition, depth of field*) is prepended to the row-level scene description. A strict anti-text negative block (*text, written text, letters, words, numbers, typography, font, readable text, handwriting, graffiti, holding sign, holding poster, holding board, holding banner, holding placard, label, sticker, badge, caption, subtitle, watermark, logo, content bubble, thought bubble*) is appended to every image's negative prompt regardless of row-level content, to prevent the model from rendering legible characters that would trivially disambiguate hate versus non-hate content. A general quality negative suffix (*low quality, worst quality, blurry, deformed, distorted, bad anatomy, cartoon, anime, oversaturated, jpeg artifacts*) follows to suppress compositional artifacts.

Alternatives Considered. Stable Diffusion XL (SDXL) and Kandinsky 3 were evaluated informally at lower scales prior to the full pipeline run. SDXL at equivalent step counts produced images with higher rates of incidental text rendering on signs and posters, increasing the risk of label leakage into the visual channel. Kandinsky 3 exhibited stronger scene-level hallucination on identity-specific

prompts, producing semantically inconsistent race-religion co-referencing. Z-Image-Turbo's combination of the Wan diffusion backbone with the Qwen3 language encoder produced more coherent grounding of identity-specific language, and its FP8 quantization made it the only model among those tested that completed the full 18,000-image corpus within available compute budget.

Leakage Audit. An independent image leakage audit confirmed that no generated image carries readable class-discriminative text after the anti-text negatives are applied. The audit report (`leakage_audit_report.json`) is included as a supplementary artifact. Checkpoint-based generation (`checkpoint\progress.csv`) further ensures that any interrupted run resumes from the last completed batch without reprocessing previously generated images, preserving per-image seed determinism throughout.